Venkata Lakshmi Parimala Pasupuleti

Staff Engineer · AI-First Engineering · Claude Code · Responsible AI · Python · React · .NET/C# · Azure · Affordable Housing Domain · 9yr Production

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PROFESSIONAL SUMMARY

Senior software engineer with 9+ years of production experience and 7 years in government-funded affordable housing technology - the exact domain Neighborly serves. At Entrata I owned the HUD income-limit and LIHTC compliance infrastructure for 7 years: the same HCV, HAP, and tax-credit workflows that Neighborly's platform administers from the government agency side. I understand this domain from the inside - eligibility calculations, income verification, federal program rules, and the consequences when compliance data is wrong. On the AI side: I am a Claude Code user daily. I built AgentProbe as a production evaluation framework with CI-integrated regression gates, LLM-as-Judge scoring, and responsible AI guardrails - exactly the kind of evaluation and quality infrastructure Neighborly needs as it scales AI use across engineering. LeaseI demonstrates production agentic AI in a regulated compliance domain with HITL guardrails where LLM reasons but deterministic logic makes every final regulated call. I have .NET Framework and C# background, React and Angular frontend experience, Azure cloud services, and a strong track record mentoring engineers and raising team-level quality through code reviews, tooling, and coaching. M.S. Data Analytics Engineering, George Mason University (GPA 3.87, May 2026).

TECHNICAL SKILLS

Languages & Frameworks: Python (9yr), C# / .NET Framework, Java, Spring Boot, PHP/Laravel, JavaScript, TypeScript, React.js, Angular, FastAPI, Flask, REST APIs, Microservices, Service-Oriented Architecture, API Design

AI-Assisted Engineering & Responsible AI: Claude Code (daily), MCP (Model Context Protocol), LangGraph, LangChain, Agentic Workflows, LLM Evaluation, LLM-as-Judge, Custom Evaluation Harnesses, CI-Integrated Regression Gates, HITL / Human-in-the-Loop, Responsible AI Guardrails, Prompting & Context Management Strategies, Hallucination Detection, RAG Pipelines, ChromaDB, Anthropic API, OpenAI API, Technical Mentoring

Cloud, Data & Infrastructure: Azure (App Service, Azure SQL, cloud services), AWS (S3, EC2, RDS, Glue, Lambda), GCP, Docker, Kubernetes, CI/CD (GitHub Actions), PostgreSQL, MySQL, Microsoft SQL Server, Oracle, NoSQL, Power BI, ETL/Data Pipelines, Data Modeling, Schema Design, Query Optimization, Observability, Structured Logging, Monitoring, Security, Compliance

EXPERIENCE

Magizhs Tech Solution

May 2025 - Aug 2025

AI/ML Engineer - AI-First Product Delivery

- Sole technical owner of a 0→1 production AI system: designed agentic pipeline architecture, built RAG retrieval layer with compliance guardrails, implemented responsible AI evaluation infrastructure and CI validation gate - delivered in weeks with measurable quality outcomes.

George Mason University

Aug 2024 - May 2026

Graduate Teaching Assistant & Research Associate

- Mentored 600+ students in Python, SQL, and distributed systems; built ML pipelines on AWS and Azure (R^2=0.977); designed Power BI and Tableau dashboards improving research comprehension 40%; applied anomaly detection across 3 research pipelines.

Entrata - Affordable Housing & Property Management SaaS

Sep 2018 - Aug 2024

Senior Software Engineer - HCV/HUD/LIHTC Compliance Systems · Lehi, UT (Remote)

- **Government affordable housing compliance - 7-year ownership** - owned HUD income-limit and LIHTC import pipeline: federal data normalization powering Housing Choice Voucher (HCV) eligibility, HAP calculations, rent determinations, and tax-credit compliance for 100K+ housing units; mentored junior engineers through annual release cycles with baseline validation and reconciliation; zero compliance violations. The property management side of the same federal programs Neighborly administers from the agency side.
- **Engineering quality & team enablement** - built internal debugging tools reducing triage time 40%; Python/Bash automation eliminating 80% of manual workflows; structured logging, monitoring, and alerting for regulated production systems; mentored 5+ engineers; 300+ code reviews; led incident response with root-cause analysis; defined and enforced coding standards across compliance-critical modules.
- **Full-stack platform delivery** - shipped 20+ production modules (Python, Java, PHP, JavaScript, React.js, Angular, PostgreSQL) across 120+ Agile sprints; Docker microservices redesign (50% scalability improvement); Kubernetes orchestration; RabbitMQ async distributed processing; GitHub Actions CI/CD; AWS S3 migration with Libsodium-encrypted ETL pipelines (100% PHI compliance).

Entrata India - Pune

Jan 2017 - Sep 2018

Associate Software Engineer → Trainee (C#/.NET · PHP · Java)

- Built affordable housing modules in C#/.NET Framework, PHP, and Java; data pipeline for 1.6M HUD/Tax Credit records; AWS S3 migration; 56 state-specific compliance document templates.

PROJECTS

AgentProbe: Responsible AI Evaluation Framework - Claude Code Powered

LLM Evaluation · Guardrails · CI Gate · Claude Code · Responsible AI · Open Source

Python · FastAPI · LangGraph · React.js · TypeScript · ChromaDB · GitHub Actions · Anthropic API · Claude Code | github.com/venkatalakshmiparimala/AgentProbe

Production AI evaluation framework built with Claude Code - demonstrating the responsible AI engineering standards Neighborly wants to establish at scale. **Evaluation infrastructure:** 5-metric quality engine (correctness, groundedness, hallucination rate, latency, token efficiency), LLM-as-Judge with pass@k multi-trial harness, CI-integrated regression gate (GitHub Actions) blocking quality degradation before deployment. **Responsible AI guardrails:** structured evaluation of AI-generated outputs, prompt/context versioning, security and quality review patterns - exactly the guardrails Neighborly needs as it scales Claude Code use across engineering teams. 15-endpoint FastAPI backend, React.js + TypeScript frontend, zero vendor lock-in across providers.

Leasel: Production Agentic AI - Regulated Affordable Housing Domain

Agentic · LangGraph · RAG · HITL · Regulated Domain · MCP · Production

Python · LangGraph (8-agent) · FastAPI · TF-IDF RAG · SQLite · MCP · Anthropic API · Claude Code | github.com/venkatalakshmiparimala/Lease_complaince_rag

Production agentic AI platform for regulated compliance - built with Claude Code, using MCP for tool integration. Direct domain analog to Neighborly's HCV/CDBG compliance workflows. **Agentic workflow:** 8-agent LangGraph StateGraph (Triage → Retrieval → Configuration → Incident → Structured Data → Reasoning → Validation → Escalation) with conditional routing and HITL escalation on low-confidence paths. **Responsible AI design:** LLM reasons and extracts; deterministic logic or human oversight makes every final regulated call - guardrail enforced in code, not convention. 93.6% precision, 87.9% F1 across 1,240 test cases with CI regression gate via AgentProbe.

AI on FHIR: Cloud-Native AI on Azure & AWS - Regulated Domain

Full Stack · Azure · AWS Lambda · HIPAA · REST APIs · Docker · PostgreSQL

Python · Flask · React.js · TypeScript · Docker · AWS Lambda · Azure · PostgreSQL · REST APIs | github.com/venkatalakshmiparimala/AI_on_FHIR

Full-stack AI application on Azure and AWS cloud infrastructure for a regulated domain - HIPAA-compliant RBAC with full audit trail, NLP-to-API query translation, Docker containerization, PostgreSQL data layer with complete observability. Demonstrates cloud-native AI development on Azure services in a compliance-first environment.

EDUCATION

M.S. Data Analytics Engineering - George Mason University, Virginia — May 2026 · GPA 3.87

B.Tech Computer Science & Engineering - JNTU Anantapur — 2013-2017